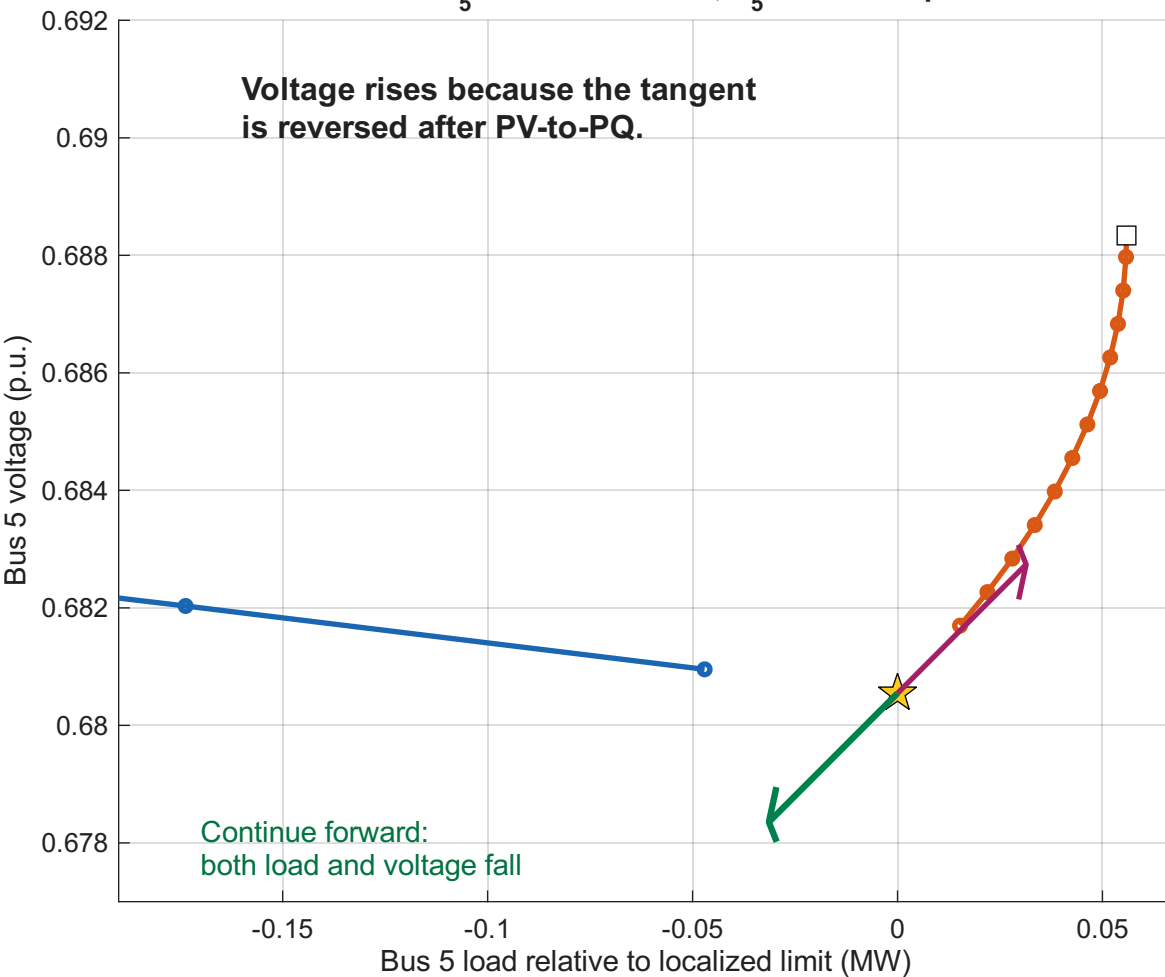


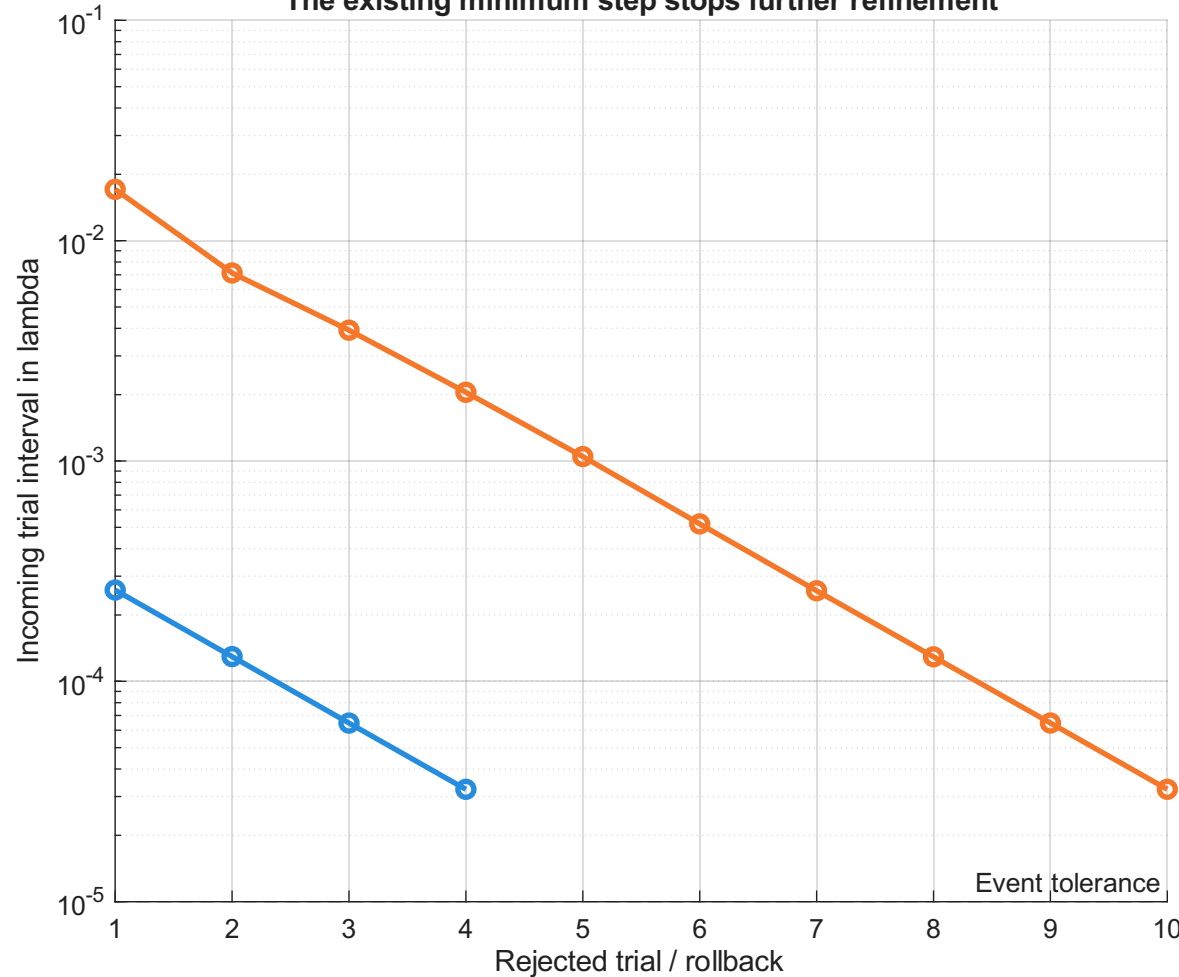
A control-limit turn can be missed even when Newton converges

Coupled current: what the upward segment means
G2 limit: $P_5 = 358.960546$ MW, $V_5 = 0.680547$ p.u.



- Original trace: G2 controls voltage
- Original trace: G2 at Q limit
- Independently localized G2 limit
- Previously reported smooth nose
- Correct outgoing tangent (local)
- Forced positive-lambda tangent

Guard detects the turn and refines the trial
The existing minimum step stops further refinement



- Coupled current, initial step 0.1
- Combined, initial step 0.1